Context Free Grammar(CFG)-based English/Urdu Translation System from Video & Audio to Pakistan Sign Language (PSL) Gestures

2024 15th International Conference on Information and Communication Systems (ICICS) | 979-8-3315-4076-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICICS63486.2024.10638303

Ahmad Bilal
Department of Computer Science,
University of Management and Technology, Lahore Pakistan
ahmadbilal227@gmail.com

Abstract— **The communication barrier between hearing impaired and unimpaired populations presents a profound challenge, impacting millions in the deaf community. Despite numerous advancements in assistive technologies, translating natural languages to Pakistan Sign Language(PSL) remains significantly underdeveloped, exacerbating communication difficulties for those with hearing impairments. This research introduces an innovative multi-stage methodology designed to bridge this gap by translating English and Urdu media content into PSL gestures using a proposed Context-free-grammar (CFG) for high grammatical precision. Our approach begins with preprocessing audio streams to ensure clarity and compatibility with subsequent processing stages. We employ the Google Speech and NLP APIs to transcribe and process media content. The identified transcription is then processed using advanced machine learning algorithms to enhance its coherence and readability, punctuating and structuring it appropriately. We then tokenize the sentences and pass it to our English PSL module which employs tools to parse and manipulate input string using PSL grammar rules. Each word of the translated sentences is matched with the HamNoSys database and the SiGML tags are generated for each symbol and then exported to the SiGML player to animate the gesture on the user interface. The system's performance was rigorously evaluated, demonstrating good reliability with an accuracy of 95.58% for sentence structures and 93.98% for tenses. The transformative potential of this work lies in enhancing accessibility and fostering inclusivity for the deaf community in Pakistan. By providing a robust real-time translation framework, our system addresses critical needs in education, healthcare, and employment.**

Keywords— Pakistan Sign Language (PSL), Context-Free-Grammar (CFG), Machine Learning, User Experience (UX), HamNoSys, Machine Translation

I. INTRODUCTION

Language serves as a principal method of human communication, consisting of words and sentences used in a structured and conventional way and conveyed by speech, writing, or gesture. [1]. In philosophy, the meaning of sentences extends beyond reference, incorporating the sense or the way the reference is presented, which is crucial in interpreting sign languages where gestures must accurately convey specific meanings [2]. Gestures and facial expressions systematically reflect linguistic and cognitive processes similar to spoken languages, providing a rich medium for conveying complex information. This equivalence underscores the potential for technological interventions to bridge communication gaps for the hearing-impaired.

Sign Language operates through a complex system of gestures and facial expressions, with variations in finger, hand, elbow, and shoulder orientations, as well as eyebrow, jaw, and lip movements, to convey nuanced meanings [3]. These gestures are further decomposed into manual gestures and non-manual gestures. Manual gestures consist of hand shape, movement, location, and orientation, whereas non-manual gestures consist of facial expression, head movement, posture and orientation of body, shoulder raising, and mouthing [4]. While spoken language integrates body gestures and facial expressions alongside verbal communication, Sign Language relies solely on body movements, particularly indispensable for individuals lacking auditory ability due to various factors such as age, accidents, or genetic predispositions [5].

According to the World Health Organization, unaddressed hearing loss globally costs $980 billion annually, encompassing healthcare, education, productivity, and societal impacts [6]. Computer science, including machine learning, computer vision, and natural language processing, holds promise for improving the lives of over 5% of the global population with disabilities [7, 8, 9]. Hearing-impaired individuals face barriers in education, employment, and services, with studies indicating lower completion rates for primary education, impacting long-term prospects [10]. In Pakistan, approximately 10 million people are deaf, with fewer than 5% of deaf children, particularly girls, attending school, contributing to high illiteracy and unemployment rates [11]. Communication challenges lead to social isolation and mental health issues, exacerbated by the lack of tools to effectively bridge spoken and sign languages, which differ significantly.

Bridging this gap is pivotal for designing a life experience based on principles of equality and inclusion, particularly as sign language serves as the primary mode of communication for over 70 million hearing-impaired individuals worldwide. Leveraging state-of-the-art technologies like Natural language Processing, Machine Learning and real-time translation offers state-of-the-art tools to develop intelligent communication systems. Investing in these technologies ensures that hearing-impaired individuals can engage fully with the world around them, addressing a moral imperative to leave no one behind in the digital age [9].

This study aims to develop a context-free grammar(CFG)-based conversion system that translates English and Urdu speech into Pakistan Sign Language (PSL), thereby addressing communication disparities faced by the hearing-impaired community in Pakistan by leveraging the state-of-the-art tools and technologies. The contribution of the study are as follows:

i. Development of a robust and innovative methodology integrating advanced technologies to accurately translate of English/Urdu media into Pakistan Sign Language (PSL) gestures. While the scope of the system is currently limited to English/Urdu, but our selected tools enable us to translate 100+ languages to PSL later.
ii. Our system can be applied in education, healthcare, employment, and government services, allowing deaf individuals to participate more fully in these areas.

Authorized licensed use limited to: University of New South Wales. Downloaded on October 25,2026 at 13:53:17 UTC from IEEE Xplore. Restrictions apply.

II. Related Work

Countries worldwide, including Pakistan, are developing sign languages to help integrate the deaf and hearing communities to communicate. Notably, research has been focused on translating English to American Sign Language (ASL) using various computational approaches. Systems like SignSpeak [10] and HamNoSys [12] have shown promising results in recognizing and translating spoken language into sign language gestures. Despite ongoing efforts, PSL is still evolving. In Pakistan, the development of systems for translating spoken languages into Pakistan Sign Language (PSL) is still in its nascent stages. Efforts have been made to create digital resources and recognition systems for PSL, but comprehensive translation systems remain underdeveloped.

Likewise, one significant project is the Pakistan Sign Language Recognition System (PSLRS), developed by the National University of Computer and Emerging Sciences (NUCES). PSLRS focuses on recognizing PSL gestures using machine learning techniques. While it provides a foundational framework for gesture recognition, it lacks the capability to translate spoken or written English/Urdu into PSL gestures comprehensively. Other key institutions contributing to its development include ABSA in Karachi, SDA in Rawalpindi, PAD in Karachi, and NISE in Islamabad [5]

NLP enables accurate translation of spoken or written text into sign language, while ML enhances gesture recognition and translation through extensive dataset learning. Transformer-based models with data augmentation and semi-supervised learning are proving effective. Real-time translation technologies, blending computer vision and NLP, provide immediate spoken-to-sign language conversion. Such as, proposed the Multi-Stream Keypoint Attention Network (MSKA-SLR) [15] to improve sign language recognition and translation by focusing on keypoint-based methods instead of traditional RGB video inputs, which are prone to background noise. This approach leverages human cognitive processes to analyze gesture configurations and supplementary elements, enhancing robustness and reducing computational demands. By integrating keypoint fusion strategies, head fusion, and self-distillation, the MSKA-SLR effectively captures keypoint sequence knowledge, achieving state-of-the-art performance on benchmarks like Phoenix-2014T.

On the other hand, technology is gradually advancing in Pakistan, with researchers striving to bridge the communication gap between the hearing and the deaf. Mehdi & Khan [6] developed "Talking Hands," a sensory glove that translates ASL into text to facilitate communication between hearing and deaf individuals. Similarly, Bukhari et al. [10] created a sensory glove that converts ASL into speech by recognizing finger movements. Similarly, Abbas and Summaira [16] focused on addressing communication barriers between the hearing and the deaf communities in Pakistan by developing a tool to translate speech and text into Pakistan Sign Language (PSL) with bilingual subtitles. This tool aims to bridge the gap in understanding and resources, particularly in educational settings where teachers struggle to communicate effectively with deaf students. Utilizing technologies like Python, Natural Language Toolkit, and pre-recorded PSL videos, they designed a prototype that facilitates better interaction and learning.

Significant progress has been made globally to bridge the communication gap between the hearing and the deaf. In Spain, Colas et al. [11] introduced a 3G mobile video calling feature that converts spoken language to sign language and vice versa. Moustakas et al. [17] created multimodal tools for communication among mute, impaired, and deaf individuals, later presented at the 2006 'Interface' summer workshop. Noberto et al. [9] in Portugal used this approach to develop a game-based program. In Tunisia, Othman & Jemni [18] translated English text to ASL using a modified Mosses device, with results displayed via a 3D avatar. A similar 3D avatar system for sign language translation was developed in 2012 by the National Technical Institute. Kumar et al. [19] also developed a tool to enable deaf individuals to produce speech sounds and communicate independently.

To the best of our knowledge and comprehensive literature review, a notable dearth of studies focuses specifically on developing a grammar-based conversion system for translating English and Urdu speech into Pakistan Sign Language (PSL). Despite significant advancements in sign language interpretation globally, the unique linguistic and cultural context of PSL remains largely unexplored, highlighting the novelty and importance of the proposed research endeavour.

III. Materials and Methods

The success of our research hinges not only on the advanced technologies, algorithms, and methodologies employed and developed but also on a mindset of commitment to excellence, rigorous verification, and a human-centered approach. Below, we outline the comprehensive processes and rationales that guided our work.

A. Data Collection and Verification

Our commitment to quality data involves sourcing from reliable and diverse origins, including primary and deaf schools, which provided stories and commonly used phrases, and recorded videos of adults and older adults speaking in English and Urdu. Additionally, recognizing the internet's role in learning and entertainment, we gathered various data types from online resources, entertainment videos, and video surveys to ensure data diversification. This approach allowed us to develop a solution capable of accepting multiple video and audio formats for comprehensive testing and verification.

B. Proposed Methodology

Our proposed methodology involves a multi-stage process to convert english/urdu content media files into grammatically correct PSL gestures. The primary stages include preprocessing, word recognition, translation and punctuation, sentence extraction, English to PSL conversion, (Steps about Hamnosys etc to be added after finalizing that section). Each stage is designed to ensure the accurate and meaningful translation of spoken language to gesture animations. The code and implementation details of the proposed system can be found at:

https://github.com/AhmadBilal227/Ahmad_Bilal_PSL_research_repo

1) Preprocessing

Reformatting the audio stream is essential for processing efficiency and accuracy, particularly for large files with unnecessary information. The process begins with MediaToolKit, which extracts audio from video sources, handling formats like .mp3, .wav, .flac for audio, and .wmv, .3gp, .mp4 for video. The preferred format is Waveform Audio (.wav) due to its compressed nature and compatibility. To meet Google Speech API's requirements (sample rate of 44100 Hz and single channel), algorithms convert audio files accordingly, including converting stereo files to mono. The Sax tool then

Authorized licensed use limited to: University of New South Wales. Downloaded on October 25,2026 at 13:53:17 UTC from IEEE Xplore. Restrictions apply.

splits the .wav file into smaller segments to comply with the API's time constraints. These segments are processed individually through the Google Cloud Speech API, which returns recognized words for further analysis.

2) Word Recognition

Extracting words from a digital audio signal involves challenges related to the speaker's phonetic accuracy and the comprehensiveness of the word dictionary. To overcome these, we use the Google Speech API, which utilizes Machine Learning and Neural Network algorithms for accurate recognition of diverse pronunciations and extensive vocabulary. Setting up the Google speech client involves obtaining an API key, enabling necessary APIs, and configuring recognition parameters. The audio file is then submitted for analysis, resulting in a string of recognized words, which is further processed to extract meaningful sentences.

3) Translation and Punctuation

Upon receiving the recognized string, which contains solely the identified words, the process of extracting meaningful sentences necessitates punctuation. For English sentences, we employ a Python script equipped with a bi-directional recurrent neural network model to restore missing punctuations, thereby enhancing the coherence and readability of the text. Conversely, for Urdu sentences, we utilize the Google Translation API to translate the string into English while preserving its intended meaning, resulting in a punctuated English string. This punctuated sentence serves as a crucial input for the subsequent module, which annotates sub-sentences for further processing and analysis.

4) Sentence Extraction

The process of sub-sentence extraction largely depends on the punctuation accuracy in the sentence. Sentences must be annotated based on sentence termination symbol i.e. period and then extracted from punctuated string.

First, we have to set up Google Language Client credentials, as we are using Google Language API, by setting up input document type to plain text and content type to text. Then we have to request annotation of our sentence in meaningful sub-sentences, which, in response, returned sentences that are stored in a list for further processing. This list of sentences are used to generate equivalent PSL sentences. This API also responds to us with parts of speech tagging and the sentiment value of the sentence, which can be used for further research in future.

5) English to PSL Conversion

Once the English phrases have been extracted from the media file and saved in a list, they are delivered to the English to Pakistan Sign Language (PSL) module over a socket connection to produce comparable PSL sentences. This component is in charge of converting the English/Urdu statement into its comparable sign language sentence. This module accepts a phrase as input and conducts morphological analysis on it before converting it into lexical units. Then it interfaces with the NL grammar component to validate sentence structure and build parse trees. This parse tree is then used to filter subcomponents of the sentence manipulator, removing stop words such as a, an, the, and other prepositions. This filtered tree serves as the input for PSL grammar, which then translates the NL filtered sentence into an equivalent PSL phrase. We discussed this conversion process briefly below:

Getting Output from Natural Language Parser

First the module uses Stanford's NLP for processing sentences to get sentence composition and identification of its constituent elements. The parser gives its output in a bracketed notation.

Sentence:
My dog is chasing a black cat.
Natural Language Parser Output:
(ROOT (S (NP (PRP$ My) (NN dog)) (VP (VBZ is) (VP (VBG chasing) (NP (DT a) (JJ black) (NN cat)))) (. .)))

Stanford's Natural Language Parser uses the grammar tags from Penn Treebank Tag-set for tagging linguistic structure of natural languages.

Saving Grammar Information in a Tree Data-Structure

To process the grammar information of English Sentences, and translating them into Pakistan Sign Language sentences. The output from Natural Language Parser is stored in a tree data-structure. For the sentence example given in previous section, the NLP output is

Natural Language Parser Output:

(ROOT (S (NP (PRP$ My) (NN dog)) (VP (VBZ is) (VP (VBG chasing) (NP (DT a) (JJ black) (NN cat)))) (. .)))

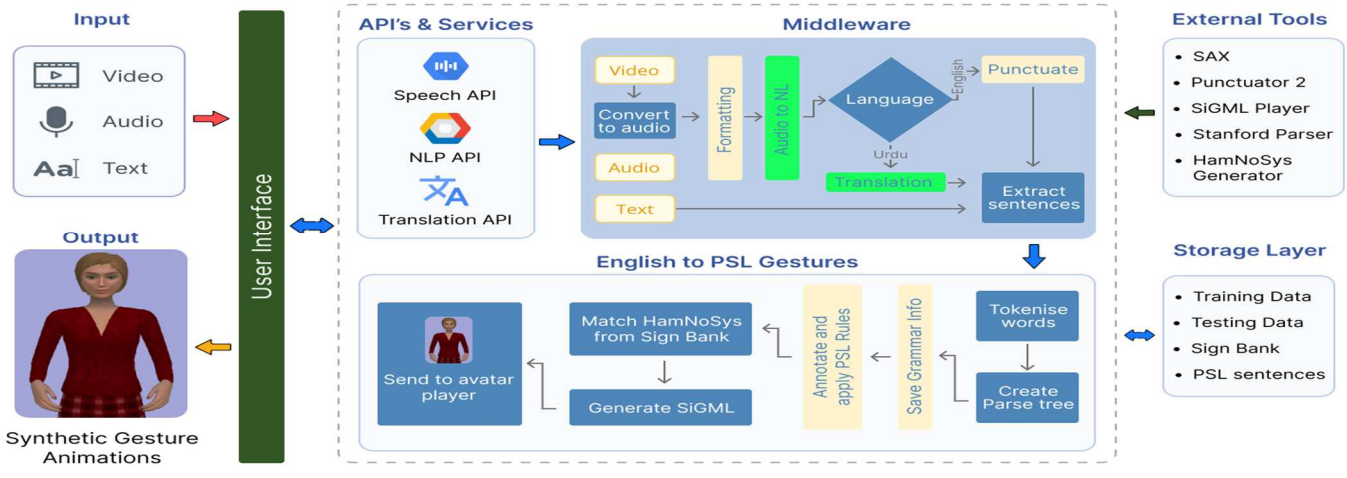


Fig 1. Architectural Diagram of Proposed System

Authorized licensed use limited to: University of New South Wales. Downloaded on October 25,2026 at 13:53:17 UTC from IEEE Xplore. Restrictions apply.

The tree data-structure constructed from this output can be viewed as given in Fig 2:

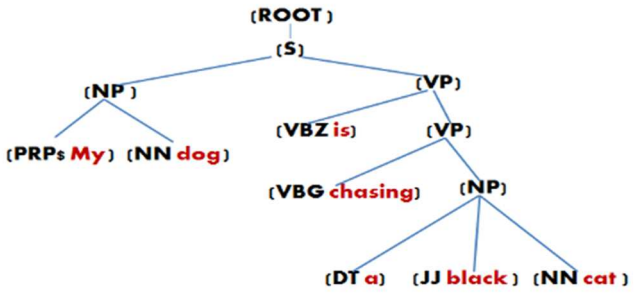


Fig 2. English Tree constructed from NLP Output

Getting Dependencies, Annotated Tree & Renaming English Tree

Another tool called Stanford's typed dependencies Tool, gives information about the dependencies between the words of input sentences. This dependency output actually represents a directed graph, where every node represents the word, and the edges represent the dependencies between the words. The tree data structure constructed from NLP output, is annotated by adding the word dependency information in it. This tree data structure now has additional information stored in word-level nodes. In this constructed English language tree, the words are tagged using Treebank Tag-set. These tags are different than the tags (names) in the grammar developed for Pakistan Sign Language, which contains a simplified and easy-to-understand tag-set.

After the identification of sentence type and tense, the Penn tree-bank tags are renamed to the PSL grammar tags using a mapping function. The Fig. 3 shows the renamed English tree.

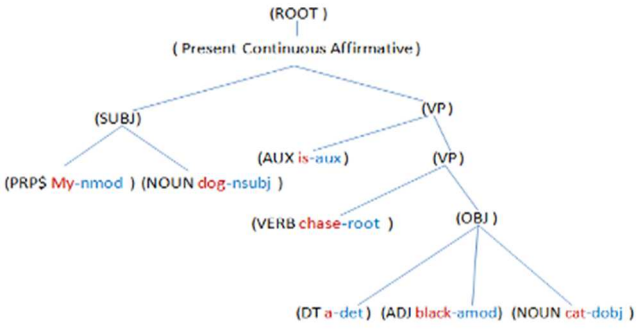


Fig 3. Output renamed English Tree

Construction of PSL Tree

Once the English sentence treeis renamed, the tree is parsed from top to bottom (pre-order traversal). PSL grammar rules are input to the program in the form of a text file. This file contains PSL grammar productions written in Backus-Naur form. At each non-terminal (non-leaf) node, the children of that node are compared by the PSL grammar productions for that non-terminal and a best-matched production is chosen. The tree is modified according to the matched PSL production. The modification operations on a non-terminal node are of three types:

1. Addition of a new child node
2. Removal of a child node
3. Re-ordering of children nodes

The new tree thus constructed has all productions according to PSL grammar. Let us take the example of the sentence used in previous sections:

```
<PRESENT-CONTINUOUS-AFFIRMATIVE> ::= <SUBJ> <VP> <"now">
<PRESENT-PERFECT-AFFIRMATIVE> ::= <SUBJ> <VP> <"full">
<PRESENT-PERFECT-CONTINUOUS-AFFIRMATIVE> ::= <SUBJ> <VP> <"full">
<PAST-INDEFINITE-AFFIRMATIVE> ::= <"was"> <SUBJ> <VP>
<PAST-CONTINUOUS-AFFIRMATIVE> ::= <"was"> <SUBJ> <VP>
<PAST-PERFECT-AFFIRMATIVE> ::= <"was"> <SUBJ> <VP> <"full">
<PAST-PERFECT-CONTINUOUS-AFFIRMATIVE> ::= <"was"> <SUBJ> <VP> <"full">
<FUTURE-INDEFINITE-AFFIRMATIVE> ::= <SUBJ> <VP> <"after">
<FUTURE-CONTINUOUS-AFFIRMATIVE> ::= <SUBJ> <VP> <"after">
<FUTURE-PERFECT-AFFIRMATIVE> ::= <SUBJ> <VP> <"full"> <"after">
<FUTURE-PERFECT-CONTINUOUS-AFFIRMATIVE> ::= <SUBJ> <VP> <"full"> <"after">

<SUBJ> ::= <NOUN> | <ADJ> <NOUN> | <PP> <NP> | <NOUN1> <NOUN2> | <NP1>
<NP2> | <NP> <UCP> | <NOUN1><CC><NOUN2> | <NP><VP> | <CD><NOUN>

<OBJ> ::= <NOUN> | <ADJ> <NOUN> | <PP> <NP> | <NOUN1> <NOUN2> | <NP1> <NP2>
| <NP> <UCP> | <NOUN1><CC><NOUN2> | <NP><VP> | <CD><NOUN>

<NP> ::= <NOUN> | <ADJ> <NOUN> | <PP> <NP> | <NOUN1> <NOUN2> | <NP1> <NP2>
| <NP> <UCP> | <NOUN1><CC><NOUN2> | <NP><VP> | <CD><NOUN>

<VP> ::= <ADVP> <VERB> | <OBJ> <VERB> | <PP> <VP> | <VERB> | <PP> <VERB> |
<ADJP> | <OBJ> <ADVP> <VERB> | <ADJP> <VERB> | <OBJ><PP><VERB> |
<VERB><PRT> | <VP1><CC><VP2>

<VP1> ::= <ADVP> <VERB> | <OBJ> <VERB> | <PP> <VP> | <VERB> | <PP> <VERB> |
<ADJP> | <OBJ> <ADVP> <VERB> | <ADJP> <VERB> | <OBJ><PP><VERB> |
```

Fig 4. PSL grammar and production rules snippet

Input Sentence:

My dog is chasing a black cat.

Grammar rule for present-continuous-affirmative tense:

<PRESENT-CONTINUOUS-AFFIRMATIVE> ::= <SUBJ> <VP> <"now">

After performing a series of these operations on this sentence's English tree to convert it to an equivalent PSL tree, the output tree can be observed in Fig. 5 and 6:

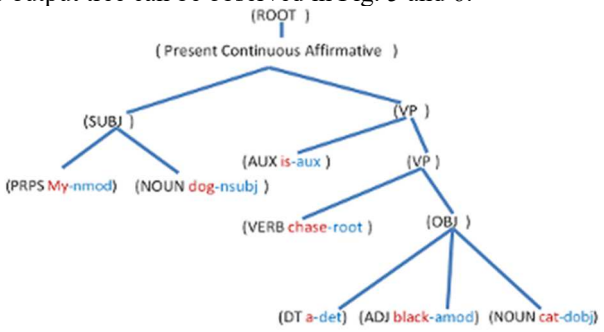


Fig. 5. Renamed English Tree

Therefore, the final PSL tree will be:

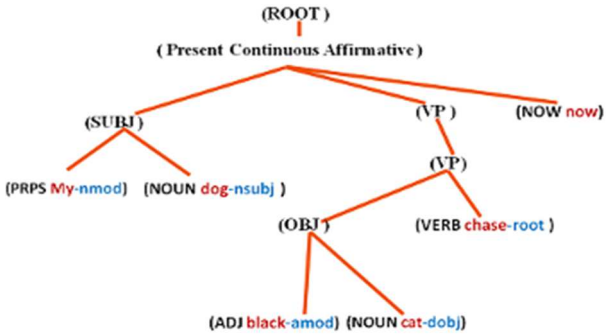


Fig. 6. Equivalent PSL Tree

Translated Sentence

The PSL tree is traversed again for a leaf-only traversal. The value (word) encountered on each leaf node is concatenated in a resultant string variable. After the traversal, we get PSL sentences in the resultant string variable.

Fig. 7. shows the order in which the leaf nodes will be visited.

Authorized licensed use limited to: University of New South Wales. Downloaded on October 25,2026 at 13:53:17 UTC from IEEE Xplore. Restrictions apply.

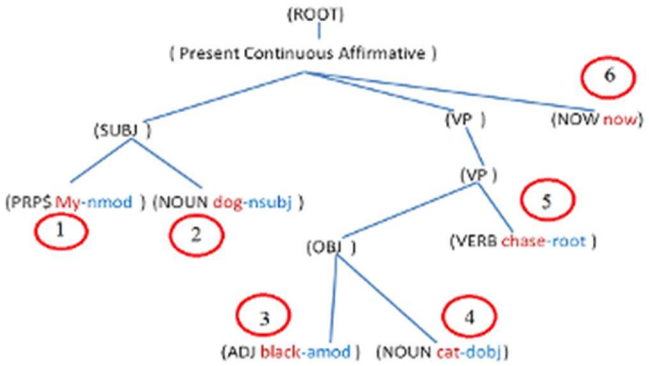


Fig. 7. Translated English Tree to PSL

The resultant PSL sentence will be:

My dog black cat chase now.

Likewise, this module takes English sentences as input and returns their equivalent PSL sentences to our main module.

6) PSL Sentence to 3D Gestures

Once we convert our English sentences to PSL sentences, these PSL sentences communicate with the Sign bank to generate HamNoSys regarding each tagged lexical unit. HamNoSys is the notation we are using to map the signs. In the end, the generated HamNoSys is converted to SigML so the external service of Avatar can take the SigML as input to generate the Sign animations of the input sentence.

SignWriting Notation

Sign Languages, like spoken languages, may be documented using SignWriting Notation Systems. Different notation systems exist for representing signs in Sign Language, however no sign language notation is now considered standard. The primary benefits of utilizing sign language notation systems are:

- They provide natural language words in a manner suitable for translation to sign language animations.
- They enhance the translation system's scalability.
- Storage space

There are several notation systems used for sign language writing, with the four most popular being Stokoe, Gloss, SignWriting, and HamNoSys. HamNoSys is versatile, not tied to any specific sign language, and can represent both manual and non-manual aspects of gestures. The notation system, created by the University of Hamburg in 1985, uses specific notations and phonetic transcriptions for defining signs and motions. HamNoSys, derived from the Stokoe notation system, provides an alphabetic method for detailing hand shape, movement, location, and orientation, making it easy to represent and process signs on a computer.

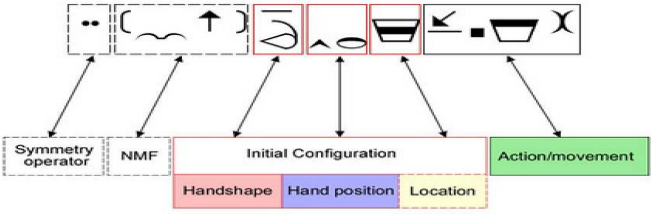


Fig. 8. HamNoSys notation composition

Sign Bank

To enable translation from text to sign language, we require a corpus of gestures for all words, as well as their HamNoSys representation. We saved a word together with all potential HamNoSys representations. In order to standardize one of the HamNoSys, we collaborated with PSL specialist and interpreters to ensure that this sign bank is acknowledged internationally by all PSL academics and developers. We also guarantee that the granularity of data units is included with the approval of PSL specialists. This standard corpus also helps us when translating from sign to text since even if a person uses a non-standard gesture during conversation, our system can map the gesture to the relevant HamNoSys. It promotes translation from sign language to regular language and vice versa.

Animating Gestures

Each word of the translated sentences is matched with the HamNoSys database and the SiGML tags are generated for each symbol and then exported to a separate file containing the SiGML. SiGML is an XML-based machine-readable code that is given to SiGML player to animate the gestures.

We integrated this third-party service into our web and cloud architecture where we can control the user experience (UX). This method proved to be effective in making all these tools accessible to the users so they can test and validate the solution.

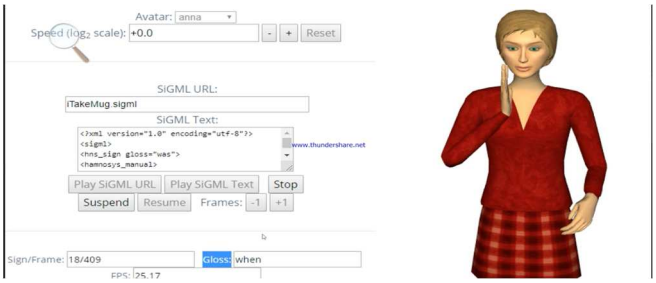


Fig. 9. User Interface prototype

IV. RESULTS AND VALIDATION

To validate the effectiveness and accuracy of our proposed solution, we conducted comprehensive tests focusing on three key aspects: Sentence Structure Type, Sentence Tense Type, and Story Recognition. We evaluated how well the system translates various grammatical structures and maintains the integrity of simple to complex sentences from English/Urdu into Pakistan Sign Language (PSL) gestures. We also assessed its ability to convey the correct tense—past, present, or future—ensuring temporal coherence in the translations. Additionally, we tested the system's performance in translating longer, narrative content to evaluate its capability to preserve logical consistency and coherence in extended discourse.

A. Sentence Recognition Accuracy

Our system demonstrated strong capabilities across various sentence structures, achieving high accuracy in converting different grammatical patterns into Pakistan Sign Language (PSL) gestures. It performed exceptionally well with Simple Sentences and Complex Sentences, achieving 95% and 95.13% accuracy respectively. Compound Sentences were translated with 92.5% accuracy, while Compound-Complex Sentences had an accuracy of 93.3%. Overall, the system maintained an impressive average accuracy of 93.98%, showcasing its robustness and flexibility in handling a wide range of syntactic structures. In terms of tense translation, the system preserved temporal consistency effectively. Future Tense sentences achieved the highest accuracy at 96.5%, followed by Present Tense sentences at 96%. Past Tense sentences had a slightly lower accuracy of 94.25% but still performed well. Overall, the system achieved an average accuracy of 95.8% in tense translation, ensuring faithful representation of the original temporal context in PSL translations.

Authorized licensed use limited to: University of New South Wales. Downloaded on October 25,2026 at 13:53:17 UTC from IEEE Xplore. Restrictions apply.

B. Story Recognition Accuracy

For story recognition, we employed audio stories that were recorded by the researchers, and audio stories downloaded from the internet. In this case, the performance of the system was determined by the ability of the system to recognize and translate entire stories into PSL gestures. Story recognition was identified to be slightly lower than that of the single sentences, with the mean score of 94.25% which is still a strong performance given the increased complexity of narrative content. This metric highlights the system's proficiency in maintaining coherence and meaning over longer texts.

TABLE III. STORY RECOGNITION ACCURACY

Story	Total Words	Correctly Recognized Words	Misinterpreted Words	Accuracy
Greedy Dog	91	90	1	98.90%
Thirsty Crow	68	61	7	89.70%
Thirsty Crow V2	157	148	9	94.26%
Clever Fox	137	135	2	98.54%

V. User Experience

The User Experience (UX) of the English/Urdu to Pakistan Sign Language (PSL) translation is easy to use, fast, and convenient. The aim is to develop a smooth experience that will improve the interaction making it easier for the deaf to use them. The translation of gestures by using avatars is advantageous over human interpreters in the following ways. The use of avatars ensure quality and are always at the interpreter's disposal unlike human interpreters. This makes it possible for the users to be able to translate at their own will and thus increases their self-sufficiency.

VI. Limitations and Future Work

It is significant to note that although the system produced great results, some limitations were identified. The quality of gesture recognition is affected by the quality of the input audio as well as the clarity of the spoken words. Further, it may fail in recognizing the gestures or translate them correctly hence resulting in a wrong interpretation of some or all the complicated or slight gestures.

Future work will focus on incorporating new language models like BERT and GPT in the TransSign CFG system. Some of these LLMs could provide features for understanding and producing natural language, which could tremendously improve the accuracy of translated sign languages and the context they are translated into.

VII. Conclusion

This research addressed the communication gap between the hearing and hearing-impaired communities in Pakistan by developing a system to improve accessibility for the deaf community, providing a robust framework for real-time translation and fostering inclusivity. Proposed system incorporates Google Speech API, HamNoSys notation and machine learning, and demonstrated a high level of accuracy in the recognition of sentence structures and tenses at around 95.58% and 93.98% respectively. The outcomes with high accuracy rates in recognition of the sentence and story prove the applicability of the proposed system for practical usage in educational, healthcare and employment fields.

References

[1] U. Farooq, M. S. M. Rahim, N. Sabir, A. Hussain, and A. Abid, "Advances in machine translation for sign language: approaches, limitations, and challenges," *Neural Comput. Appl.*, vol. 33, no. 21, pp. 14357–14399, 2021.

[2] S. Gaukroger, "Literature : The Relevance of Logic , Language and Frege," vol. 6, no. 1, pp. 68–96, 1983.

[3] B. Hassan, M. S. Farooq, A. Abid, and N. Sabir, "Pakistan Sign Language: Computer Vision Analysis & Recommendations," *VFAST Trans. Softw. Eng.*, vol. 9, no. 1, p. 1, 2015, doi: 10.21015/vtse.v9i1.386.

[4] R. Elliott and J. R. Kennaway, "A Framework for Non-Manual Gestures in a Synthetic Signing System," *Language (Baltim).*, no. August, 1991.

[5] N. S. Khan, A. Abid, K. Abid, U. Farooq, M. S. Farooq, and H. Jameel, "Speak Pakistan : Challenges in Developing Pakistan Sign," *South Asian Stud. A Res. J. South Asian Stud.*, vol. 30, no. 2, pp. 367–379, 2015, [Online]. Available: http://journals.pu.edu.pk/journals/index.php/IJSAS/article/view/3027.

[6] S. A. Mehdi and E. Sciences, "Sign Language Recognition using Sensor Gloves Yasir Niaz Khan," *Neural Inf. Process.*, vol. 5, pp. 2204–2206, 2007.

[7] S. Nuthakki, C. S. Kulkarni, S. Kathiriya, and Y. Nuthakki, "Artificial Intelligence Applications in Natural Gas Industry: A Literature Review," *Int. J. Eng. Adv. Technol.*, vol. 13, no. 3, pp. 64–70, 2024, doi: 10.35940/ijeat.c4383.13030224.

[8] S. Nuthakki, "CONVERSATIONAL AI AND LLM ' S CURRENT AND FUTURE IMPACTS IN IMPROVING AND," no. December, 2023, doi: 10.17605/OSF.IO/7GHJY.

[9] U. Farooq, M. S. Mohd Rahim, and A. Abid, "A multi-stack RNN-based neural machine translation model for English to Pakistan sign language translation," *Neural Comput. Appl.*, vol. 35, no. 18, pp. 13225–13238, 2023, doi: 10.1007/s00521-023-08424-0.

[10] J. Bukhari, M. Rehman, S. I. Malik, A. M. Kamboh, and A. Salman, "American Sign Language Translation through Sensory Glove; SignSpeak," *Int. J. u- e-Service, Sci. Technol.*, vol. 8, no. 1, pp. 131–142, 2015, doi: 10.14257/ijunesst.2015.8.1.12.

[11] J. Tejedor, F. López, D. Bolaños, and J. Colás, "Augmented Service for Deaf People Using a Text to Sign Language Translator," 2010, [Online]. Available: http://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.551.9529.

[12] T. Hanke, "HamNoSys - Representing sign language data in language resources and language processing contexts," *Lr. 2004, Work. Proc. Represent. Process. sign Lang.*, no. June, pp. 1–6, 2004, [Online]. Available: http://www.sign-lang.uni-hamburg.de/dgs-korpus/files/inhalt_pdf/HankeLRECSLP2004_05.pdf.

[13] H. Plus, "Honyaku Plus," 2024. https://www.honyaku-plus.com/en/japanese-english-translation/ (accessed Jul. 07, 2024).

[14] ProDeaf, "ProDeaf," 2024. https://prodeaf-libras-translator.soft112.com/#google_vignette (accessed Sep. 07, 2024).

[15] M. Guan, Y. Wang, G. Ma, J. Liu, and M. Sun, "Multi-Stream Keypoint Attention Network for Sign Language Recognition and Translation," pp. 1–15, 2024, [Online]. Available: http://arxiv.org/abs/2405.05672.

[16] A. Abbas and S. Sarfraz, "Developing a Prototype to Translate Text and Speech to Pakistan Sign Language With Bilingual Subtitles: A Framework," *J. Educ. Technol. Syst.*, vol. 47, no. 2, pp. 248–266, 2018, doi: 10.1177/0047239518794168.

[17] K. Moustakas, G. Nikolakis, D. Tzovaras, B. Deville, I. Marras, and J. Pavlek, "Multimodal Tools and Interfaces for the Intercommunication between Visually Impaired and Deaf and Mute People," *eNTERFACE 2006 Publ. Proc.*, 2006.

[18] A. Othman and M. Jemni, "Statistical Sign Language Machine Translation: from English written text to American Sign Language Gloss," vol. 8, no. 5, pp. 65–73, 2011, [Online]. Available: http://arxiv.org/abs/1112.0168.

[19] S. K. S, J. J, N. I, and S. A, "An Embedded Module as 'VIRTUAL TONGUE' for Voiceless," *Int. J. Inf. Sci. Tech.*, vol. 4, no. 3, pp. 155–163, 2014, doi: 10.5121/ijist.2014.4319.



Ahmad Bilal is a principal designer and researcher with a distinguished background in Software Engineering and a fervent advocate for inclusive design. He holds a Bachelor's degree in Software Engineering and is currently advancing his expertise by applying state-of-the-art methodologies to address real, impactful issues. His area of expertise are user experience design, natural language processing, and machine learning

Authorized licensed use limited to: University of New South Wales. Downloaded on October 25,2026 at 13:53:17 UTC from IEEE Xplore. Restrictions apply.